Question ID:161
Topic Name: Mathematics-Section A

Question: Let $A = \{x \in \mathbb{R} : |x+1| < 2\}$ and $B = \{x \in \mathbb{R} : |x-1| \geq 2\}$. Then which one of the following statements is **NOT** true ?

A $A - B = (-1, 1)$

B $B - A = \mathbb{R} - (-3, 1)$

C $A \cap B = (-3, -1]$

D $A \cup B = \mathbb{R} - [1, 3)$

Answer Given By Candidate:C

Question ID:162
Topic Name: Mathematics-Section A

Question: Let $a, b \in \mathbb{R}$ be such that the equation $ax^2 - 2bx + 15 = 0$ has a repeated root α. If α and β are the roots of the equation $x^2 - 2bx + 21 = 0$, then $\alpha^2 + \beta^2$ is equal to:

A 37

B 58

C 68

D 92

Answer Given By Candidate:Not Attempted

Question ID:163
Topic Name: Mathematics-Section A

Question: Let z_1 and z_2 be two complex numbers such that $\bar{z}_1 = i\bar{z}_2$ and $\arg\left(\frac{z_1}{\bar{z}_2}\right) = \pi$. Then

A $\arg z_2 = \frac{\pi}{4}$

B $\arg z_2 = -\frac{3\pi}{4}$

C $\arg z_1 = \frac{\pi}{4}$

D $\arg z_1 = -\frac{3\pi}{4}$

Answer Given By Candidate:**Not Attempted**

Question ID:164
Topic Name: Mathematics-Section A

Question: The system of equations

$$-kx + 3y - 14z = 25$$

$$-15x + 4y - kz = 3$$

$$-4x + y + 3z = 4$$

is consistent for all k in the set

A $\mathbb{R}$

B $\mathbb{R} - \{-11, 13\}$

C $\mathbb{R} - \{13\}$

D $\mathbb{R} - \{-11, 11\}$

Answer Given By Candidate:**D**

Question ID:165
Topic Name: Mathematics-Section A

Question: $\lim\limits_{x \to \frac{\pi}{2}} \left(\tan^2 x \left(\left(2\sin^2 x + 3\sin x + 4\right)^{\frac{1}{2}} - \left(\sin^2 x + 6\sin x + 2\right)^{\frac{1}{2}} \right) \right)$ is equal to

A $\frac{1}{12}$

B $-\frac{1}{18}$

C $-\frac{1}{12}$

D $\frac{1}{6}$

Answer Given By Candidate:**Not Attempted**

Question ID:166
Topic Name: Mathematics-Section A

Question: The area of the region enclosed between the parabolas $y^2 = 2x - 1$ and $y^2 = 4x - 3$ is

A $\frac{1}{3}$

B $\frac{1}{6}$

C

$\frac{2}{3}$

D $\frac{3}{4}$

Answer Given By Candidate: **Not Attempted**

Question ID: 167
Topic Name: Mathematics-Section A

Question: The coefficient of x^{101} in the expression $(5+x)^{500} + x(5+x)^{499} + x^2(5+x)^{498} + \ldots\ldots + x^{500}$, $x > 0$, is

A ${}^{501}C_{101}\,(5)^{399}$

B ${}^{501}C_{101}\,(5)^{400}$

C ${}^{501}C_{100}\,(5)^{400}$

D ${}^{500}C_{101}\,(5)^{399}$

Answer Given By Candidate: **Not Attempted**

Question ID: 168
Topic Name: Mathematics-Section A

Question: The sum $1 + 2 \cdot 3 + 3 \cdot 3^2 + \ldots\ldots + 10 \cdot 3^9$ is equal to :

A $\frac{2 \cdot 3^{12} + 10}{4}$

B $\frac{19 \cdot 3^{10} + 1}{4}$

C $5 \cdot 3^{10} - 2$

D $\frac{9 \cdot 3^{10} + 1}{2}$

Answer Given By Candidate: **Not Attempted**

Question ID: 169
Topic Name: Mathematics-Section A

Question: Let P be the plane passing through the intersection of the planes $\vec{r} \cdot \left(\hat{i} + 3\hat{j} - \hat{k}\right) = 5$ and $\vec{r} \cdot \left(2\hat{i} - \hat{j} + \hat{k}\right) = 3$, and the point $(2, 1, -2)$. Let the position vectors of the points X and Y be $\hat{i} - 2\hat{j} + 4\hat{k}$ and $5\hat{i} - \hat{j} + 2\hat{k}$ respectively. Then the points

A X and $X + Y$ are on the same side of P

B Y and $Y - X$ are on the opposite sides of P

C

X and Y are on the opposite sides of P

D $X+Y$ and $X-Y$ are on the same side of P

Answer Given By Candidate:**Not Attempted**

Question ID:1610
Topic Name: Mathematics-Section A
Question: A circle touches both the y-axis and the line $x+y=0$. Then the locus of its center is:

A $y=\sqrt{2}x$

B $x=\sqrt{2}y$

C $y^2-x^2=2xy$

D $x^2-y^2=2xy$

Answer Given By Candidate:**Not Attempted**

Question ID:1611
Topic Name: Mathematics-Section A
Question: Water is being filled at the rate of 1 cm^3 / sec in a right circular conical vessel (vertex downwards) of height 35 cm and diameter 14 cm. When the height of the water level is 10 cm, the rate (in cm^2 / sec) at which the wet conical surface area of the vessel increases is

A 5

B $\frac{\sqrt{21}}{5}$

C $\frac{\sqrt{26}}{5}$

D $\frac{\sqrt{26}}{10}$

Answer Given By Candidate:**Not Attempted**

Question ID:1612
Topic Name: Mathematics-Section A
Question: If $b_n=\int_0^{\frac{\pi}{2}}\frac{\cos^2 nx}{\sin x}dx,\ n\in\mathbb{N}$, then

A $b_3-b_2, b_4-b_3, b_5-b_4$ are in an A.P. with common difference –2

B $\frac{1}{b_3-b_2}, \frac{1}{b_4-b_3}, \frac{1}{b_5-b_4}$ are in an A.P. with common difference 2

C $b_3-b_2, b_4-b_3, b_5-b_4$ are in a G.P.

D $\frac{1}{b_3-b_2}, \frac{1}{b_4-b_3}, \frac{1}{b_5-b_4}$ are in an A.P. with common difference –2

Answer Given By Candidate: **Not Attempted**

Question ID: 1613
Topic Name: Mathematics-Section A
Question: If $y = y(x)$ is the solution of the differential equation $2x^2 \frac{dy}{dx} - 2xy + 3y^2 = 0$ such that $y(e) = \frac{e}{3}$, then $y(1)$ is equal to

A $\frac{1}{3}$

B $\frac{2}{3}$

C $\frac{3}{2}$

D 3

Answer Given By Candidate: **B**

Question ID: 1614
Topic Name: Mathematics-Section A
Question: If the angle made by the tangent at the point (x_0, y_0) on the curve
$x = 12\,(t + \sin t \cos t)$,
$y = 12\,(1 + \sin t)^2$, $0 < t < \frac{\pi}{2}$, with the positive x-axis is $\frac{\pi}{3}$, then y_0 is equal to:

A $6\left(3+2\sqrt{2}\right)$

B $3\left(7+4\sqrt{3}\right)$

C 27

D 48

Answer Given By Candidate: **Not Attempted**

Question ID: 1615
Topic Name: Mathematics-Section A
Question: The value of $2\sin(12°) - \sin(72°)$ is :

A $\frac{\sqrt{5}\left(1-\sqrt{3}\right)}{4}$

B $\frac{1-\sqrt{5}}{8}$

C

$\frac{\sqrt{3}\left(1-\sqrt{5}\right)}{2}$

D $\frac{\sqrt{3}\left(1-\sqrt{5}\right)}{4}$

Answer Given By Candidate: **Not Attempted**

Question ID: 1616
Topic Name: Mathematics-Section A

Question: A biased die is marked with numbers 2, 4, 8, 16, 32, 32 on its faces and the probability of getting a face with mark n is $\frac{1}{n}$. If the die is thrown thrice, then the probability, that the sum of the numbers obtained is 48, is :

A $\frac{7}{2^{11}}$

B $\frac{7}{2^{12}}$

C $\frac{3}{2^{10}}$

D $\frac{13}{2^{12}}$

Answer Given By Candidate: **Not Attempted**

Question ID: 1617
Topic Name: Mathematics-Section A

Question: The negation of the Boolean expression $((\sim q) \wedge p) \Rightarrow ((\sim p) \vee q)$ is logically equivalent to :

A $p \Rightarrow q$

B $q \Rightarrow p$

C $\sim(p \Rightarrow q)$

D $\sim(q \Rightarrow p)$

Answer Given By Candidate: **A**

Question ID: 1618
Topic Name: Mathematics-Section A

Question: If the line $y = 4 + kx$, $k > 0$, is the tangent to the parabola $y = x - x^2$ at the point P and V is the vertex of the parabola, then the slope of the line through P and V is:

A $\frac{3}{2}$

B $\frac{26}{9}$

C $\frac{5}{2}$

D $\frac{23}{6}$

Answer Given By Candidate:**Not Attempted**

Question ID:1619
Topic Name: Mathematics-Section A

Question: The value of $\tan^{-1}\left(\frac{\cos\left(\frac{15\pi}{4}\right)-1}{\sin\left(\frac{\pi}{4}\right)}\right)$ is equal to :

A $-\frac{\pi}{4}$

B $-\frac{\pi}{8}$

C $-\frac{5\pi}{12}$

D $-\frac{4\pi}{9}$

Answer Given By Candidate:**Not Attempted**

Question ID:1620
Topic Name: Mathematics-Section A

Question: The line $y = x + 1$ meets the ellipse $\frac{x^2}{4}+\frac{y^2}{2}=1$ at two points P and Q. If r is the radius of the circle with PQ as diameter then $(3r)^2$ is equal to :

A 20

B 12

C 11

D 8

Answer Given By Candidate:**Not Attempted**

Question ID:1621
Topic Name: Mathematics-Section B

Question: Let $A=\begin{pmatrix}2 & -2\\ 1 & -1\end{pmatrix}$ and $B=\begin{pmatrix}-1 & 2\\ -1 & 2\end{pmatrix}$. Then the number of elements in the set $\{(n, m) : n, m \in \{1, 2,.....,10\}$ and $nA^n + mB^m = I\}$ is _____.

Answer Given By Candidate:**Not Attempted**

Question ID:1622

Topic Name: Mathematics-Section B

Question: Let $f(x)=\left[2x^2+1\right]$ and $g(x)=\begin{cases}2x-3, & x<0\\ 2x+3, & x\geq 0\end{cases}$, where $[t]$ is the greatest integer $\leq t$. Then, in the open interval $(-1, 1)$, the number of points where fog is discontinuous is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:1623

Topic Name: Mathematics-Section B

Question: The value of $b>3$ for which $12\int_3^b \frac{1}{(x^2-1)(x^2-4)}dx=\log_e\left(\frac{49}{40}\right)$, is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:1624

Topic Name: Mathematics-Section B

Question: If the sum of the co-efficients of all the positive even powers of x in the binomial expansion of $\left(2x^3+\frac{3}{x}\right)^{10}$ is $5^{10}-\beta\cdot 3^9$, then β is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:1625

Topic Name: Mathematics-Section B

Question: If the mean deviation about the mean of the numbers 1, 2, 3, ….., n, where n is odd, is $\frac{5(n+1)}{n}$, then n is equal to _____.

Answer Given By Candidate:**10**

Question ID:1626

Topic Name: Mathematics-Section B

Question: Let $\vec{b}=\hat{i}+\hat{j}+\lambda\hat{k}, \lambda\in\mathbb{R}$. If $\vec{a}$ is a vector such that $\vec{a}\times\vec{b}=13\hat{i}-\hat{j}-4\hat{k}$ and $\vec{a}\cdot\vec{b}+21=0$, then $\left(\vec{b}-\vec{a}\right)\cdot\left(\hat{k}-\hat{j}\right)+\left(\vec{b}+\vec{a}\right)\cdot\left(\hat{i}-\hat{k}\right)$ is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:1627

Topic Name: Mathematics-Section B

Question: The total number of three-digit numbers, with one digit repeated exactly two times, is _____.

Answer Given By Candidate:**Not Attempted**

Question ID:1628

Topic Name: Mathematics-Section B

Question: Let $f(x) = |(x-1)(x^2-2x-3)| + x - 3$, $x \in \mathbb{R}$. If m and M are respectively the number of points of local minimum and local maximum of f in the interval $(0, 4)$, then $m + M$ is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:1629
Topic Name: Mathematics-Section B

Question: Let the eccentricity of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ be $\frac{5}{4}$. If the equation of the normal at the point $\left(\frac{8}{\sqrt{5}}, \frac{12}{5}\right)$ on the hyperbola is $8\sqrt{5}x + \beta y = \lambda$, then $\lambda - \beta$ is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:1630
Topic Name: Mathematics-Section B

Question: Let l_1 be the line in xy-plane with x and y intercepts $\frac{1}{8}$ and $\frac{1}{4\sqrt{2}}$ respectively, and l_2 be the line in zx-plane with x and z intercepts $-\frac{1}{8}$ and $-\frac{1}{6\sqrt{3}}$ respectively. If d is the shortest distance between the line l_1 and l_2, then d^{-2} is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:1631
Topic Name: Physics-Section A

Question: Given below are two statements. One is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A: Two identical balls A and B thrown with same velocity 'u' at two different angles with horizontal attained the same range R. If A and B reached the maximum height h_1 and h_2 respectively, then $R = 4\sqrt{h_1 h_2}$

Reason R: Product of said heights.

$h_1 h_2 = \left(\frac{u^2 \sin^2\theta}{2g}\right) \cdot \left(\frac{u^2 \cos^2\theta}{2g}\right)$

Choose the correct answer :

A Both **A** and **R** are true and **R** is the correct explanation of **A**.

B Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.

C **A** is true but **R** is false.

D **A** is false but **R** is true.

Answer Given By Candidate:**Not Attempted**

Question ID:1632
Topic Name: Physics-Section A

Question: Two buses P and Q start from a point at the same time and move in a straight line and their positions are represented by $X_P(t) = \alpha t + \beta t^2$ and $X_Q(t) = ft - t^2$. At what time, both the buses have same velocity ?

A $\frac{\alpha - f}{1+\beta}$

B $\frac{\alpha + f}{2(\beta - 1)}$

C $\frac{\alpha + f}{2(1+\beta)}$

D $\frac{f - \alpha}{2(1+\beta)}$

Answer Given By Candidate:**D**

Question ID:1633
Topic Name: Physics-Section A

Question: A disc with a flat small bottom beaker placed on it at a distance R from its center is revolving about an axis passing through the center and perpendicular to its plane with an angular velocity ω. The coefficient of static friction between the bottom of the beaker and the surface of the disc is μ. The beaker will revolve with the disc if :

A $R \leq \frac{\mu g}{2\omega^2}$

B $R \leq \frac{\mu g}{\omega^2}$

C $R \geq \frac{\mu g}{2\omega^2}$

D $R \geq \frac{\mu g}{\omega^2}$

Answer Given By Candidate:**Not Attempted**

Question ID:1634
Topic Name: Physics-Section A

Question: A solid metallic cube having total surface area 24 m^2 is uniformly heated. If its temperature is increased by 10°C, calculate the increase in volume of the cube. (Given $\alpha = 5.0 \times 10^{-4}\ °C^{-1}$) .

A $2.4 \times 10^6\ cm^3$

B $1.2 \times 10^5\ cm^3$

C $6.0 \times 10^4\ cm^3$

D $4.8 \times 10^5\ cm^3$

Answer Given By Candidate:**A**

Question ID:1635
Topic Name: Physics-Section A

Question: A copper block of mass 5.0 kg is heated to a temperature of 500°C and is placed on a large ice block. What is the maximum amount of ice that can melt? [Specific heat of copper : 0.39 J g^{-1} °C^{-1} and latent heat of fusion of water : 335 J g^{-1}]

A 1.5 kg

B 5.8 kg

C 2.9 kg

D 3.8 kg

Answer Given By Candidate:**Not Attempted**

Question ID:1636
Topic Name: Physics-Section A

Question: The ratio of specific heats $\left(\frac{C_P}{C_V}\right)$ in terms of degree of freedom (f) is given by :

A $\left(1+\frac{f}{3}\right)$

B $\left(1+\frac{2}{f}\right)$

C $\left(1+\frac{f}{2}\right)$

D $\left(1+\frac{1}{f}\right)$

Answer Given By Candidate:**B**

Question ID:1637
Topic Name: Physics-Section A

Question: For a particle in uniform circular motion, the acceleration $\vec{a}$ at any point $P(R,\theta)$ on the circular path of radius R *is (when* θ is measured from the positive x-axis and v is uniform speed) :

A $-\frac{v^2}{R}\sin\theta\,\hat{i}+\frac{v^2}{R}\cos\theta\,\hat{j}$

B $-\frac{v^2}{R}\cos\theta\,\hat{i}+\frac{v^2}{R}\sin\theta\,\hat{j}$

C $-\frac{v^2}{R}\cos\theta\,\hat{i}-\frac{v^2}{R}\sin\theta\,\hat{j}$

D $-\frac{v^2}{R}\hat{i}+\frac{v^2}{R}\hat{j}$

Answer Given By Candidate:**C**

Question ID:1638
Topic Name: Physics-Section A

Question: Two metallic plates form a parallel plate capacitor. The distance between the plates is 'd'. A metal sheet of thickness $\frac{d}{2}$ and of area equal to area of each plate is introduced between the plates. What will be the ratio of the new capacitance to the original capacitance of the capacitor?

A 2:1

B 1:2

C 1:4

D 4:1

Answer Given By Candidate:**A**

Question ID:1639
Topic Name: Physics-Section A

Question: Two cells of same emf but different internal resistances r_1 and r_2 are connected in series with a resistance R. The value of resistance R, for which the potential difference across second cell is zero, is :

A $r_2 - r_1$

B $r_1 - r_2$

C r_1

D r_2

Answer Given By Candidate:**Not Attempted**

Question ID:1640
Topic Name: Physics-Section A

Question: Given below are two statements :
Statement – I : Susceptibilities of paramagnetic and ferromagnetic substances increase with decrease in temperature.
Statement – II : Diamagnetism is a result of orbital motions of electrons developing magnetic moments opposite to the applied magnetic field.
Choose the correct answer from the options given below :-

A Both Statement – I and Statement – II are true.

B Both Statement – I and Statement – II are false.

C Statement – I is true but Statement – II is false.

D Statement – I is false but Statement – II is true.

Answer Given By Candidate:**Not Attempted**

Question ID:1641
Topic Name: Physics-Section A

Question: A long solenoid carrying a current produces a magnetic field B along its axis. If the current is doubled and the number of turns per cm is halved, the new value of magnetic field will be equal to

A B

B $2B$

C $4B$

D $\frac{B}{2}$

Answer Given By Candidate:**A**

Question ID:1642
Topic Name: Physics-Section A

Question: A sinusoidal voltage $V(t) = 210 \sin 3000\, t$ volt is applied to a series LCR circuit in which $L = 10\ mH$, $C = 25\ \mu F$ and $R = 100\ \Omega$. The phase difference (Φ)between the applied voltage and resultant current will be :

A $\tan^{-1}(0.17)$

B $\tan^{-1}(9.46)$

C $\tan^{-1}(0.30)$

D $\tan^{-1}(13.33)$

Answer Given By Candidate:**Not Attempted**

Question ID:1643
Topic Name: Physics-Section A

Question: The electromagnetic waves travel in a medium at a speed of 2.0×10^8 m/s. The relative permeability of the medium is 1.0. The relative permittivity of the medium will be :

A 2.25

B 4.25

C 6.25

D 8.25

Answer Given By Candidate:**Not Attempted**

Question ID:1644
Topic Name: Physics-Section A

Question: The interference pattern is obtained with two coherent light sources of intensity ratio 4:1. And the ratio $\frac{I_{max} + I_{min}}{I_{max} - I_{min}}$ is $\frac{5}{x}$. Then, the value of x will be equal to :

A 3

B 4

C 2

D 1

Answer Given By Candidate: **B**

Question ID: 1645
Topic Name: Physics-Section A

Question: A light whose electric field vectors are completely removed by using a good polaroid, allowed to incident on the surface of the prism at Brewster's angle. Choose the most suitable option for the phenomenon related to the prism.

A Reflected and refracted rays will be perpendicular to each other.

B Wave will propagate along the surface of prism.

C No refraction, and there will be total reflection of light.

D No reflection. and there will be total transmission of light.

Answer Given By Candidate: **A**

Question ID: 1646
Topic Name: Physics-Section A

Question: A proton, a neutron, an electron and an α-particle have same energy. If λ_p, λ_n, λ_e and λ_α are the de Broglie's wavelengths of proton, neutron, electron and α particle respectively, then choose the correct relation from the following :

A $\lambda_p = \lambda_n > \lambda_e > \lambda_\alpha$

B $\lambda_\alpha < \lambda_n < \lambda_p < \lambda_e$

C $\lambda_e < \lambda_p = \lambda_n > \lambda_\alpha$

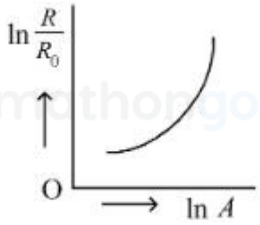
D $\lambda_e = \lambda_p = \lambda_n = \lambda_\alpha$

Answer Given By Candidate: **Not Attempted**

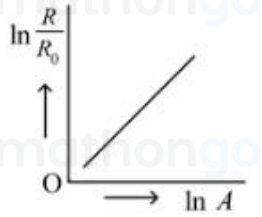
Question ID: 1647
Topic Name: Physics-Section A

Question: Which of the following figure represents the variation of $\ln\left(\frac{R}{R_0}\right)$ with $\ln A$ (if R = radius of a nucleus and A = its mass number)

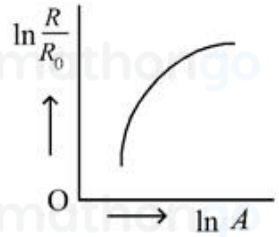
A



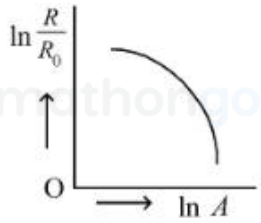
B



C



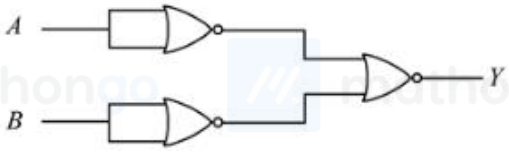
D



Answer Given By Candidate: **B**

Question ID: 1648
Topic Name: Physics-Section A

Identify the logic operation performed by the given circuit :



Question:

A AND gate

B OR gate

C NOR gate

D NAND gate

Answer Given By Candidate: **A**

Question ID: 1649
Topic Name: Physics-Section A

Match List I with List II

List I	List II
A. Facsimile	I. Static Document Image
B. Guided media Channel	II. Local Broadcast Radio
C. Frequency Modulation	III. Rectangular wave
D. Digital Signal	IV. Optical Fiber

Question: Choose the correct answer from the following options :

A A-IV, B-III, C-II, D-I

B A-I, B-IV, C-II, D-III

C A-IV, B-II, C-III, D-I

D A-I, B-II, C-III, D-IV

Answer Given By Candidate: **D**

Question ID: 1650
Topic Name: Physics-Section A

Question: If n represents the actual number of deflections in a converted galvanometer of resistance G and shunt resistance S. Then the total current I when its figure of merit is K will be :

A $\frac{KS}{(S+G)}$

B $\frac{(G+S)}{nKS}$

C $\frac{nKS}{(G+S)}$

D $\frac{nK(G+S)}{S}$

Answer Given By Candidate: **Not Attempted**

Question ID: 1651
Topic Name: Physics-Section B

Question: For $z = a^2x^3y^{\frac{1}{2}}$, where '$a$' is a constant. If percentage error in measurement of 'x' and 'y' are 4% and 12%, respectively, then the percentage error for 'z' will be _____ %.

Answer Given By Candidate: **18**

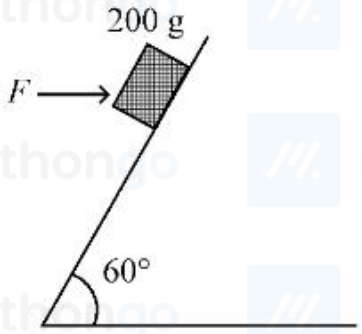
Question ID: 1652
Topic Name: Physics-Section B

Question: A curved in a level road has a radius 75 m. The maximum speed of a car turning this curved road can be 30 m/s without skidding. If radius of curved road is changed to 48 m and the coefficient of friction between the tyres and the road remains same, then maximum allowed speed would be _____ m/s.

Answer Given By Candidate: **Not Attempted**

Question ID: 1653
Topic Name: Physics-Section B

A block of mass 200 g is kept stationary on a smooth inclined plane by applying a minimum horizontal force $F = \sqrt{x}N$ as shown in figure.



Question: The value of x = _____.

Answer Given By Candidate: **12**

Question ID: 1654
Topic Name: Physics-Section B

Moment of Inertia (M.I.) of four bodies having same mass 'M' and radius '$2R$' are as follows :

I_1 = M.I. of solid sphere about its diameter

I_2 = M.I. of solid cylinder about its axis

I_3 = M.I. of solid circular disc about its diameter

I_4 = M.I. of thin circular ring about its diameter

Question: If $2(I_2 + I_3) + I_4 = x \cdot I_1$ then the value of x will be _____.

Answer Given By Candidate: **Not Attempted**

Question ID: 1655
Topic Name: Physics-Section B

Two satellites S_1 and S_2 are revolving in circular orbits around a planet with radius $R_1 = 3200\ km$ and $R_2 = 800\ km$ respectively. The ratio of speed of satellite S_1 to the speed of satellite S_2 in their respective orbits would be $\frac{1}{x}$ where x =

Question: _____.

Answer Given By Candidate: **4**

Question ID: 1656
Topic Name: Physics-Section B

When a gas filled in a closed vessel is heated by raising the temperature by 1°C, its pressure increases by 0.4%. The initial temperature of the gas is _____ K.

Question:

Answer Given By Candidate: **Not Attempted**

Question ID: 1657
Topic Name: Physics-Section B

27 identical drops are charged at $22V$ each. They combine to form a bigger drop. The potential of the bigger drop will be _____ V.

Question:

Answer Given By Candidate:**Not Attempted**

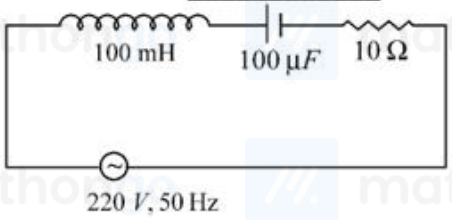
Question ID:1658
Topic Name: Physics-Section B

Question: The length of a given cylindrical wire is increased to double of its original length. The percentage increase in the resistance of the wire will be _____%.

Answer Given By Candidate:**100**

Question ID:1659
Topic Name: Physics-Section B

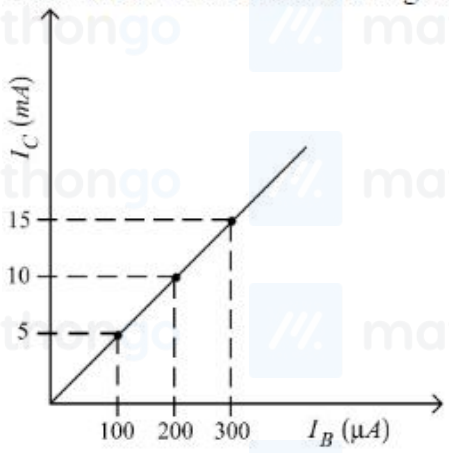
Question: In a series LCR circuit, the inductance, capacitance and resistance are L = 100 mH, C= 100 μF and $R=10\ \Omega$ respectively. They are connected to an AC source of voltage 220 V and frequency of 50 Hz. The approximate value of current in the circuit will be _____ A.



Answer Given By Candidate:**Not Attempted**

Question ID:1660
Topic Name: Physics-Section B

Question: In an experiment of CE configuration of n-p-n transistor, the transfer characteristics are observed as given in figure.



If the input resistance is 200 Ω and output resistance is 60 Ω, the voltage gain in this experiment will be _____.

Answer Given By Candidate:**15**

Question ID:1661
Topic Name: Chemistry-Section A

Question: The minimum energy that must be possessed by photons in order to produce the photoelectric effect with platinum metal is:
[Given: The threshold frequency of platinum is $1.3 \times 10^{15}\ s^{-1}$ and $h = 6.6 \times 10^{-34}$ J s.]

A 3.21×10^{-14} J

B 6.24×10^{-16} J

C 8.58×10^{-19} J

D 9.76×10^{-20} J

Answer Given By Candidate: C

Question ID: 1662
Topic Name: Chemistry-Section A

Question: At 25°C and 1 atm pressure, the enthalpy of combustion of benzene (l) and acetylene (g) are -3268 kJ mol^{-1} and -1300 kJ mol^{-1}, respectively. The change in enthalpy for the reaction $3\ C_2H_2(g) \rightarrow C_6H_6$ (l), is

A +324 kJ mol^{-1}

B +632 kJ mol^{-1}

C -632 kJ mol^{-1}

D -732 kJ mol^{-1}

Answer Given By Candidate: C

Question ID: 1663
Topic Name: Chemistry-Section A

Question: Solute A associates in water. When 0.7 g of solute A is dissolved in 42.0 g of water, it depresses the freezing point by 0.2 °C. The percentage association of solute A in water, is :
[Given: Molar mass of A = 93 g mol^{-1}. Molal depression constant of water is 1.86 K kg mol^{-1}.]

A 50%

B 60%

C 70%

D 80%

Answer Given By Candidate: Not Attempted

Question ID: 1664
Topic Name: Chemistry-Section A

Question: The K_{sp} for bismuth sulphide (Bi_2S_3) is 1.08×10^{-73}. The solubility of Bi_2S_3 in mol L^{-1} at 298 K is

A 1.0×10^{-15}

B 2.7×10^{-12}

C 3.2×10^{-10}

D 4.2×10^{-8}

Answer Given By Candidate: **A**

Question ID: 1665
Topic Name: Chemistry-Section A

Match **List I** with **List II**.

List I	List II
A. Zymase	I. Stomach
B. Diastase	II. Yeast
C. Urease	III. Malt
D. Pepsin	IV. Soyabean

Question: Choose the correct answer from the options given below:

A A-II, B-III, C-I, D-IV

B A-II, B-III, C-IV, D-I

C A-III, B-II, C-IV, D-I

D A-III, B-II, C-I, D-IV

Answer Given By Candidate: **C**

Question ID: 1666
Topic Name: Chemistry-Section A

Question: The correct order of electron gain enthalpies of Cl, F, Te and Po is

A $F < Cl < Te < Po$

B $Po < Te < F < Cl$

C $Te < Po < Cl < F$

D $Cl < F < Te < Po$

Answer Given By Candidate: **B**

Question ID: 1667
Topic Name: Chemistry-Section A

Given below are two statements.

Statement I: During electrolytic refining, blister copper deposits precious metals.
Statement II: In the process of obtaining pure copper by electrolysis method, copper blister is used to make the anode.

Question: In the light of the above statements, choose the *correct* answer from the options given below.

A Both Statement I and Statement II are **true**.

B Both Statement I and Statement II are **false**.

C Statement I is **true** but Statement II is **false**.

D Statement I is **false** but Statement II is **true**.

Answer Given By Candidate: **Not Attempted**

Question ID: 1668
Topic Name: Chemistry-Section A

Given below are two statements one is labelled as **Assertion A** and the other is labelled as **Reason R** :

Assertion A : The amphoteric nature of water is explained by using Lewis acid/base concept.
Reason R : Water acts as an acid with NH_3 and as a base with H_2S.

In the light of the above statements choose the ***correct*** answer from the options given below :

Question:

A Both **A** and **R** are true and **R** is the correct explanation of **A.**

B Both **A** and **R** are true but **R** is **NOT** the correct explanation of **A.**

C **A** is true but **R** is false.

D **A** is false but **R** is true.

Answer Given By Candidate: **Not Attempted**

Question ID: 1669
Topic Name: Chemistry-Section A

The correct order of reduction potentials of the following pairs is

A. Cl_2/Cl^-

B. I_2/I^-

C. Ag^+/Ag

D. Na^+/Na

E. Li^+/Li

Question: Choose the correct answer from the options given below.

A A > C > B > D > E

B A > B > C > D > E

C A > C > B > E > D

D A > B > C > E > D

Answer Given By Candidate: **B**

Question ID: 1670
Topic Name: Chemistry-Section A

The number of bridged oxygen atoms present in compound B formed from the following reactions is

$$Pb(NO_3)_2 \xrightarrow{673\ K} A + PbO + O_2$$

Question: $A \xrightarrow{\text{Dimerise}} B$

A 0

B 1

C 2

D 3

Answer Given By Candidate: **A**

Question ID: 1671
Topic Name: Chemistry-Section A
Question: The metal ion (in gaseous state) with lowest spin-only magnetic moment value is

A V^{2+}

B Ni^{2+}

C Cr^{2+}

D Fe^{2+}

Answer Given By Candidate: **B**

Question ID: 1672
Topic Name: Chemistry-Section A

Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A: Polluted water may have a value of BOD of the order of 17 ppm.
Reason R: BOD is a measure of oxygen required to oxidise both the bio-degradable and non-biodegradable organic material in water.

Question: In the light of the above statements, choose the ***most appropriate*** answer from the options given below.

A Both **A** and **R** are correct and **R** is the correct explanation of **A.**

B Both **A** and **R** are correct but **R** is NOT the correct explanation of **A.**

C **A** is correct but **R** is not correct.

D **A** is not correct but **R** is correct.

Answer Given By Candidate: **C**

Question ID: 1673
Topic Name: Chemistry-Section A

Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A: A mixture contains benzoic acid and napthalene. The pure benzoic acid can be separated out by the use of benzene.
Reason R: Benzoic acid is soluble in hot water.

Question: In the light of the above statements, choose the ***most appropriate*** answer from the options given below.

A Both **A** and **R** are true and **R** is the correct explanation of **A.**

B Both **A** and **R** are true but **R** is NOT the correct explanation of **A.**

C **A** is true but **R** is false.

D **A** is false but **R** is true.

Answer Given By Candidate: **Not Attempted**

Question ID: 1674
Topic Name: Chemistry-Section A

Question: During halogen test, sodium fusion extract is boiled with concentrated HNO_3 to

A remove unreacted sodium

B decompose cyanide or sulphide of sodium

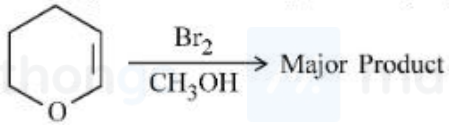
C extract halogen from organic compound

D maintain the pH of extract.

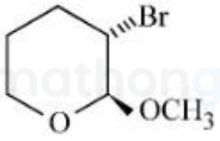
Answer Given By Candidate: **B**

Question ID: 1675
Topic Name: Chemistry-Section A

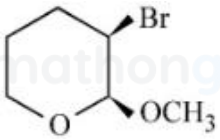
Question: Amongst the following, the major product of the given chemical reaction is



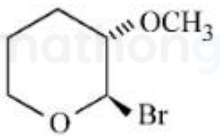
A



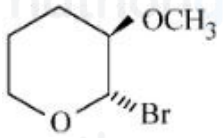
B



C



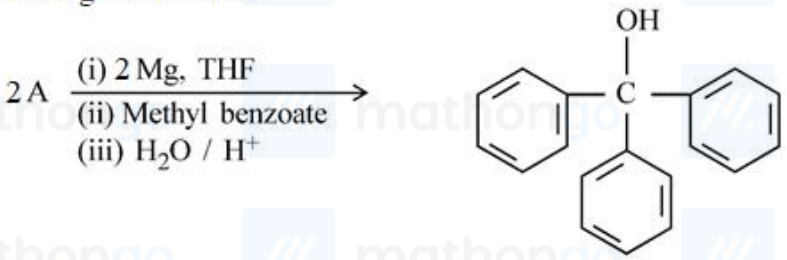
D



Answer Given By Candidate: **A**

Question ID: 1676
Topic Name: Chemistry-Section A

In the given reaction



Question: 'A' can be

A benzyl bromide

B bromobenzene

C cyclohexyl bromide

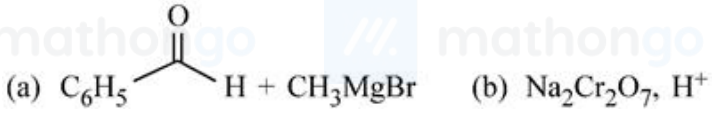
D methyl bromide

Answer Given By Candidate: **B**

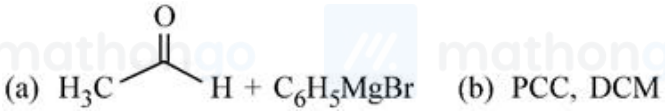
Question ID: 1677
Topic Name: Chemistry-Section A

Question: Which of the following conditions or reaction sequence will **NOT** give acetophenone as the major product ?

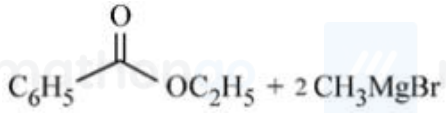
A



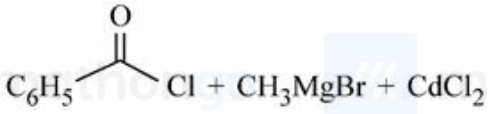
B



C



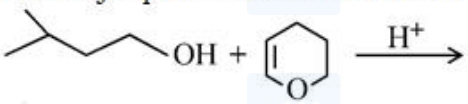
D



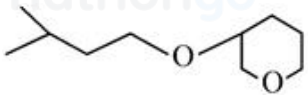
Answer Given By Candidate: **Not Attempted**

Question ID: 1678
Topic Name: Chemistry-Section A

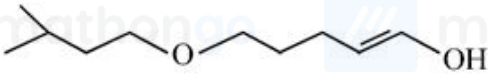
Question: The major product formed in the following reaction, is



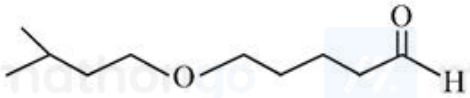
A



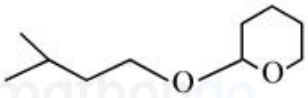
B



C



D

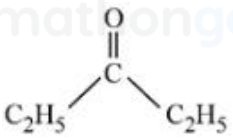


Answer Given By Candidate: **Not Attempted**

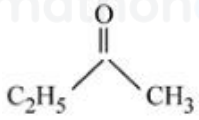
Question ID: 1679
Topic Name: Chemistry-Section A

Question: Which of the following ketone will **NOT** give enamine on treatment with secondary amines? [where t-Bu is $-C(CH_3)_3$]

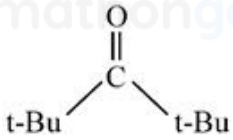
A



B



C



D



Answer Given By Candidate: **Not Attempted**

Question ID: 1680
Topic Name: Chemistry-Section A

Question: An antiseptic dettol is a mixture of two compounds 'A' and 'B' where A has 6π electrons and B has 2π electrons. What is 'B'?

A Bithionol

B Terpineol

C Chloroxylenol

D Chloramphenicol

Answer Given By Candidate:**B**

Question ID:1681
Topic Name: Chemistry-Section B
Question: A protein 'A' contains 0.30% of glycine (molecular weight 75). The minimum molar mass of the protein 'A' is _____ $\times 10^3$ g mol^{-1} [nearest integer]

Answer Given By Candidate:**Not Attempted**

Question ID:1682
Topic Name: Chemistry-Section B
Question: A rigid nitrogen tank stored inside a laboratory has a pressure of 30 atm at 06:00 am when the temperature is 27 °C. At 03:00 pm, when the temperature is 45°C, the pressure in the tank will be _____ atm. [nearest integer]

Answer Given By Candidate:**32**

Question ID:1683
Topic Name: Chemistry-Section B
Question: Amongst BeF_2, BF_3, H_2O, NH_3, CCl_4 and HCl, the number of molecules with non-zero net dipole moment is _____.

Answer Given By Candidate:**3**

Question ID:1684
Topic Name: Chemistry-Section B
Question: At 345 K, the half life for the decomposition of a sample of a gaseous compound initially at 55.5 kPa was 340 s. When the pressure was 27.8 kPa, the half life was found to be 170 s. The order of the reaction is _____. [integer answer]

Answer Given By Candidate:**Not Attempted**

Question ID:1685
Topic Name: Chemistry-Section B
Question: A solution of $Fe_2(SO_4)_3$ is electrolyzed for 'x' min with a current of 1.5 A to deposit 0.3482 g of Fe. The value of x is _____. [nearest integer]

Given : 1 F = 96500 C mol^{-1}

Atomic mass of Fe = 56 g mol^{-1}

Answer Given By Candidate:**Not Attempted**

Question ID:1686
Topic Name: Chemistry-Section B

Consider the following reactions :

$PCl_3 + H_2O \rightarrow A + HCl$

$A + H_2O \rightarrow B + HCl$

Question: The number of ionisable protons present in the product B is _____.

Answer Given By Candidate:**3**

Question ID:1687
Topic Name: Chemistry-Section B

Question: Amongst $FeCl_3.3H_2O$, $K_3[Fe(CN)_6]$ and $[Co(NH_3)_6]Cl_3$, the spin-only magnetic moment value of the inner-orbital complex that absorbs light at shortest wavelength is _____ B.M. [nearest integer]

Answer Given By Candidate:**Not Attempted**

Question ID:1688
Topic Name: Chemistry-Section B

Question: The Novolac polymer has mass of 963 g. The number of monomer units present in it are

Answer Given By Candidate:**Not Attempted**

Question ID:1689
Topic Name: Chemistry-Section B

Question: How many of the given compounds will give a positive Biuret test _____ ?
Glycine, Glycylalanine, Tripeptide, Biuret

Answer Given By Candidate:**Not Attempted**

Question ID:1690
Topic Name: Chemistry-Section B

Question: The neutralization occurs when 10 mL of 0.1M acid 'A' is allowed to react with 30 mL of 0.05 M base $M(OH)_2$. The basicity of the acid 'A' is _____.
[M is a metal]

Answer Given By Candidate:**3**